Fig. 4: Acoustic image of smaller component in Fig. 2

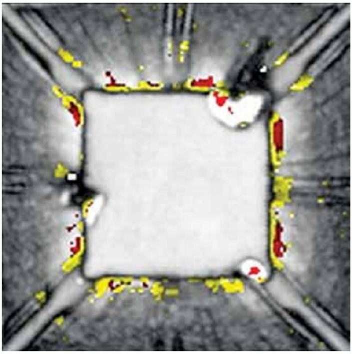

Fig. 5: Acoustic image of a larger component

tures at the corners of the die; white regions are similar but a bit more highly reflective. Both features are air-filled delaminations between the overlying mould compound and the surface of the die. A delamination on a die face, even though it may be extremely thin, has two unfortunate characteristics: It is very likely to grow in area as a result of thermal cycling over time, and it can easily shear off wire bonds that it encounters as it grows.

The die region of the larger IC package is shown in Fig. 4. This image was gated on the die surface—meaning that only echoes whose arrival time indicated that they had originated within a vertically narrow gate encompassing the die surface were used. Gates can be made very thin, if needed, and are used to avoid depth confusion among features, as well as to determine the precise depth of a single interface.

Fig. 4 displays numerous yellow, red, and white features, which are mostly associated with the perimeter of the die—meaning that the features are close to the vulnerable wire bonds. All of the anomalies can be classified as delaminations or voids. Red and white indicate the highest degree of reflection, while yellow indicates somewhat lower reflection, probably because the top surface of the anomaly is slanted rather than flat and thus directs some of the echo away from the transducer.

Some of the peripheral features may be harmless, but there are three red-on-white features on the die itself that are not. These are the one at the lower right corner of the die, the one at the upper right corner, and the one about a third of the way up the left margin of the die. Each of these is in contact with the die surface and is capable of expanding its area and cleaving wire bonds—or may already have done so.

The large defect at the upper right corner appears to have a large, somewhat indistinct black area attached to it. This is almost certainly an extension of the void (which may actually be a crack) upward toward the surface of the mould compound. The transducer is collecting only those echoes whose arrival time matches the narrow gate enclosing the die surface. If this extension of the defect extends upward above the top of the gate, the x-y region corresponding to the extension will send no return echoes and system software will fill that region with black pixels, meaning no signal received from within this gate.

The defect along the left edge of the die has a similar black region associated with it, and probably of similar origin.

One important post-reflow evaluation of mounted components uses ultrasound to measure rather than to image. Suppose a new lot of a particular component is being mounted—a component that has a solid history of few failures. The new lot, however, has abundant defects, and production halts. The acoustic micro imaging tool can measure the acoustic impedance of the mould compound on the new components and on samples of previous components. If there are significant differences between the two, it often means that the composition of the mould compound, and hence its response to the stresses of handling and reflow, has been altered by the manufacturer.

The images shown here were made with C-mode imaging, where the amplitude and polarity of each returning echo is assigned a colour. Numerous other C-SAM imaging modes can be used on mounted components to make, for example, a nondestructive cross-section through a component. Internal features, including defects, can also be imaged by two types of acoustic 3D imaging modes. The fact that a component is mounted on a board does not necessarily reduce its ability to be imaged with ultrasound. EFY

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